

Introduction to Cloud Computing and AWS

Study Guide

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UNIT 1: INTRODUCTION TO CLOUD COMPUTING

1.1 What is Cloud Computing?

Cloud Computing is the on-demand delivery of IT resources — computing power, storage, databases, networking, software, analytics, and more — over the Internet ("the cloud") with pay-as-you-go pricing.

 **Simple Definition:** Instead of buying, owning, and maintaining physical data centers and servers, you can access technology services, such as computing power, storage, and databases, on an as-needed basis from a cloud provider like AWS, Microsoft Azure, or Google Cloud.

Formal Definition (NIST)

The National Institute of Standards and Technology (NIST) defines cloud computing as:

 **NIST Definition:** "Cloud computing is a model for enabling ubiquitous, convenient, on-demand network access to a shared pool of configurable computing resources (e.g., networks, servers, storage, applications, and services) that can be rapidly provisioned and released with minimal management effort or service provider interaction."

Five Essential Characteristics (NIST)

- **On-demand self-service:** Users can provision computing capabilities automatically without human interaction with the service provider.
- **Broad network access:** Services are available over the network and accessed through standard mechanisms (laptops, mobile phones, tablets).
- **Resource pooling:** Provider's resources are pooled to serve multiple consumers (multi-tenancy) with different physical and virtual resources dynamically assigned.
- **Rapid elasticity:** Capabilities can be scaled rapidly outward and inward — appear unlimited to the consumer.
- **Measured service:** Resource usage can be monitored, controlled, and reported (pay-per-use model).

Evolution of Cloud Computing

- – John McCarthy proposed utility computing concept.
- – Salesforce.com launched first SaaS CRM application.
- – Amazon Web Services launched.
- – AWS launched EC2 (Elastic Compute Cloud) & S3 (Simple Storage Service).
- – Google App Engine, Microsoft Azure announced.

- – Cloud adoption exploded across enterprises globally.
- – Hybrid cloud, serverless, edge computing become mainstream.

UNIT 2: KEY BENEFITS OF CLOUD COMPUTING

2.1 Six Key Advantages of Cloud Computing

#	Benefit	Description
1	Trade Capital Expense for Variable Expense	Instead of investing in data centers and servers before you know how you'll use them, pay only when you consume computing resources, and pay only for how much you consume.
2	Benefit from Massive Economies of Scale	By using cloud computing, you can achieve a lower variable cost than you can get on your own. Hundreds of thousands of customers aggregate in the cloud, resulting in higher economies of scale and lower pay-as-you-go prices.
3	Stop Guessing Capacity	Eliminate guessing on your infrastructure capacity needs. When you make a capacity decision before deploying an application, you often either sit on expensive idle resources or deal with limited capacity.
4	Increase Speed and Agility	New IT resources are only a click away, reducing the time to make those resources available to developers from weeks to minutes. This results in a dramatic increase in agility.
5	Stop Spending Money on Running and Maintaining Data Centers	Focus on projects that differentiate your business, not the infrastructure. Cloud lets you focus on your customers, rather than on the heavy lifting of racking, stacking, and powering servers.
6	Go Global in Minutes	Deploy your application in multiple AWS Regions around the world with just a few clicks. This means you can provide lower latency and a better experience for your customers at minimal cost.

2.2 Additional Technical Benefits

Scalability

- **Vertical Scaling (Scaling Up):** Increasing the size of an existing resource (e.g., adding more CPU/RAM to a server).
- **Horizontal Scaling (Scaling Out):** Adding more instances of resources (e.g., adding more servers).

- **Auto Scaling:** Automatically adjusts resources based on demand — scale out during peak, scale in during low traffic.

Elasticity

- Ability to automatically acquire resources when needed and release when not needed.
- Helps organizations avoid over-provisioning and under-provisioning.

High Availability & Fault Tolerance

- **High Availability:** System continues to operate even if some components fail.
- **Fault Tolerance:** System operates properly even in the presence of hardware/software faults.
- **Disaster Recovery:** Policies and procedures to restore systems after catastrophic failure.

Security Benefits

- Cloud providers invest heavily in physical security, network security, and compliance certifications.
- Data encryption at rest and in transit.
- Identity & Access Management (IAM) controls.
- Compliance with standards: HIPAA, PCI-DSS, SOC 2, ISO 27001, GDPR.

UNIT 3: CLOUD SERVICE MODELS

3.1 Overview of Service Models

There are three primary cloud service models — IaaS, PaaS, and SaaS — each differing in the level of control and responsibility handed to the customer.

Feature	IaaS	PaaS	SaaS
Full Form	Infrastructure as a Service	Platform as a Service	Software as a Service
Who Manages Hardware?	Cloud Provider	Cloud Provider	Cloud Provider
Who Manages OS?	Customer	Cloud Provider	Cloud Provider
Who Manages Runtime?	Customer	Cloud Provider	Cloud Provider
Who Manages App?	Customer	Customer	Cloud Provider
Who Manages Data?	Customer	Customer	Cloud Provider
Control Level	Highest	Medium	Lowest

Feature	IaaS	PaaS	SaaS
Flexibility	Highest	Medium	Lowest
Ease of Use	Complex	Medium	Easiest
AWS Examples	EC2, S3, VPC	Elastic Beanstalk, RDS	Amazon WorkMail, Chime
Other Examples	Azure VM, Google CE	Heroku, Google App Engine	Gmail, Office 365, Salesforce

3.2 Infrastructure as a Service (IaaS)

Definition

IaaS provides the basic building blocks for cloud IT. It provides access to networking features, computers (virtual or on dedicated hardware), and data storage space with the highest level of flexibility and management control over IT resources.

Key Characteristics

- **Virtualized computing resources:** Servers, storage, networking delivered over the internet.
- **Pay-per-use pricing:** Billed by hour/minute/second for resources consumed.
- **Self-service provisioning:** Users spin up/down resources without manual provider involvement.
- **Scalable:** Resources can be scaled up/down rapidly.

Use Cases

- Website hosting and web apps
- Storage, backup, and recovery
- High-performance computing (HPC)
- Big data analysis
- Development and testing environments

3.3 Platform as a Service (PaaS)

Definition

PaaS removes the need to manage the underlying infrastructure and lets users focus on deployment and management of their applications. It provides a platform including hardware, OS, runtime environment, and middleware.

Key Characteristics

- **Application development focus:** Developers focus on writing code, not managing servers.
- **Built-in scalability:** Platform handles scaling automatically.

- **Integrated DevOps tools:** CI/CD pipelines, version control, testing frameworks.
- **Database management:** Managed database services (no DBA tasks for patching/backup).

Use Cases

- Application development and testing
- API development
- Business intelligence and analytics
- Database development and management

3.4 Software as a Service (SaaS)

Definition

SaaS provides a complete product that is run and managed by the service provider. With SaaS, you don't have to think about how the service is maintained or how the underlying infrastructure is managed.

Key Characteristics

- **Subscription-based:** Pay monthly/yearly license fees.
- **Accessible via browser:** No local installation required.
- **Automatic updates:** Provider handles all maintenance and upgrades.
- **Multi-tenancy:** Multiple customers share the same application instance.

Use Cases

- Email and collaboration (Gmail, Outlook 365)
- CRM (Salesforce, HubSpot)
- ERP systems
- Communication tools (Zoom, Slack, Teams)

 **Analogy:** Memory Trick — Think of Pizza as a Service: IaaS = You make the pizza yourself using the oven (infrastructure). PaaS = Restaurant provides kitchen and ingredients; you cook. SaaS = You order a ready-made pizza; just eat it!

UNIT 4: CLOUD DEPLOYMENT MODELS

4.1 Types of Cloud Deployment Models

Feature	Public Cloud	Private Cloud	Hybrid Cloud	Community Cloud
Ownership	Cloud Provider (AWS, Azure, GCP)	Single Organization	Mix of both	Shared by multiple orgs
Access	Open to all via internet	Restricted to one org	Controlled access	Group of organizations
Cost	Low (pay-per-use)	High (own hardware)	Medium	Shared costs
Scalability	Very High	Limited	High	Medium
Security Control	Provider-managed	Full control	Partial control	Shared control
Examples	AWS, Azure, Google Cloud	OpenStack, VMware vSphere	AWS Outposts, Azure Arc	Healthcare cloud, Gov cloud
Best For	Startups, variable workloads	Sensitive data, compliance	Regulated industries	Collaborative organizations

4.2 Public Cloud

- Owned and operated by third-party cloud service providers.
- Services delivered over the public internet and available to anyone.
- Hardware, software, and supporting infrastructure owned and managed by the provider.
- **Advantages:** No capital expenditure, unlimited scalability, high reliability, global reach.
- **Disadvantages:** Less control over security, possible latency, data sovereignty issues.

4.3 Private Cloud

- Cloud computing resources used exclusively by a single business or organization.
- Can be physically located on the company's on-site data center or hosted by a third party.
- Services and infrastructure maintained on a private network.
- **Advantages:** Greater control, better security, compliance-friendly, customizable.
- **Disadvantages:** High capital cost, IT expertise required, limited scalability.

4.4 Hybrid Cloud

- Combines public and private clouds, bound together by technology that allows data/applications to be shared.
- Gives businesses greater flexibility, more deployment options, and optimizes existing infrastructure, security, and compliance.
- **Use Case Example:** Keep sensitive customer data in private cloud while running web-facing applications on public cloud.
- **Advantages:** Best of both worlds, flexibility, cost optimization, compliance support.
- **Disadvantages:** Complex to set up and manage, security challenges at the integration point.

4.5 Community Cloud

- Shared infrastructure for a specific community with common concerns (security, compliance, jurisdiction).
- May be owned, managed, and operated by one or more organizations in the community, a third party, or a combination.
- **Examples:** Healthcare organizations sharing a HIPAA-compliant cloud, Government agencies.

UNIT 5: WHAT IS AWS?

5.1 Introduction to Amazon Web Services

Amazon Web Services (AWS) is the world's most comprehensive and broadly adopted cloud platform, offering over 200 fully featured services from data centers globally. Millions of customers — including the fastest-growing startups, largest enterprises, and leading government agencies — are using AWS to lower costs, become more agile, and innovate faster.

 **History:** AWS was launched in 2006, starting with Amazon S3 (storage) and Amazon EC2 (compute). It was born from Amazon's need to handle massive e-commerce infrastructure, and they realized they could offer this infrastructure to other businesses.

5.2 Key Facts About AWS

Attribute	Detail
Founded	2006 (Amazon S3 and EC2 launched)
Parent Company	Amazon.com, Inc.

Attribute	Detail
Market Position	#1 cloud provider globally (over 32% market share as of 2024)
Number of Services	200+ services covering compute, storage, database, AI/ML, IoT, security, and more
Regions	30+ geographic regions worldwide
Availability Zones	96+ Availability Zones
Edge Locations	400+ Points of Presence (PoPs)
Pricing Model	Pay-as-you-go, Reserved, Spot, Savings Plans
Free Tier	12 months free tier for new accounts + always free services

5.3 AWS Service Categories

Compute

- **Amazon EC2:** Virtual machines in the cloud (IaaS)
- **AWS Lambda:** Serverless computing — run code without provisioning servers
- **Amazon ECS/EKS:** Container management services (Docker, Kubernetes)
- **AWS Elastic Beanstalk:** PaaS — deploy web applications without managing infrastructure

Storage

- **Amazon S3:** Object storage — store and retrieve any amount of data
- **Amazon EBS:** Block storage — persistent storage for EC2 instances
- **Amazon EFS:** Elastic File System — scalable file storage
- **Amazon Glacier:** Low-cost archival storage

Database

- **Amazon RDS:** Managed relational databases (MySQL, PostgreSQL, Oracle, SQL Server, MariaDB, Aurora)
- **Amazon DynamoDB:** Fully managed NoSQL database
- **Amazon ElastiCache:** In-memory caching (Redis, Memcached)
- **Amazon Redshift:** Petabyte-scale data warehouse

Networking

- **Amazon VPC:** Virtual Private Cloud — isolated virtual network
- **Amazon Route 53:** DNS and traffic routing service
- **Amazon CloudFront:** Content Delivery Network (CDN)
- **AWS Direct Connect:** Dedicated network connection from on-premises to AWS

Security & Identity

- **AWS IAM:** Identity and Access Management — control who can access what
- **AWS Shield:** DDoS protection
- **AWS WAF:** Web Application Firewall
- **AWS KMS:** Key Management Service for encryption
- **Amazon GuardDuty:** Threat detection and monitoring

5.4 AWS Shared Responsibility Model

The Shared Responsibility Model defines what AWS is responsible for (security OF the cloud) and what the customer is responsible for (security IN the cloud).

AWS Responsible For	Customer Responsible For
Physical security of data centers	IAM policies and user management
Hardware maintenance (servers, network)	Operating system configuration and patching
Hypervisor and virtualization layer	Application security
Global network infrastructure	Data encryption and access control
Managed service security (RDS engine patches)	Network traffic configuration (Security Groups, NACLs)

UNIT 6: AWS GLOBAL INFRASTRUCTURE

6.1 Overview

The AWS Global Infrastructure is the backbone of AWS services. It is designed around Regions and Availability Zones (AZs), providing reliable, low-latency, and highly available cloud services globally.

6.2 AWS Regions

Definition

An AWS Region is a physical location in the world where AWS clusters data centers. Each Region is completely independent of other Regions.

Key Facts

- As of 2024, AWS has 30+ geographic Regions worldwide.
- Each Region has a unique name like: us-east-1 (N. Virginia), eu-west-1 (Ireland), ap-south-1 (Mumbai).
- Regions are isolated from each other for maximum fault tolerance and stability.
- Not all services are available in all Regions.

How to Choose a Region? (4 Factors)

1. **Compliance:** Data governance and legal requirements — data must stay within a country.
2. **Proximity:** Select the Region closest to customers to reduce latency.
3. **Feature Availability:** Some services are only available in certain Regions.
4. **Pricing:** Service prices vary by Region.

6.3 Availability Zones (AZs)

Definition

An Availability Zone (AZ) is one or more discrete data centers within a Region, each with redundant power, networking, and connectivity.

Key Facts

- Each AWS Region has minimum 2 and typically 3+ Availability Zones.
- AZs are physically separated within a Region (different buildings, power grids, flood plains).
- They are connected through high-bandwidth, low-latency, private fiber-optic networking.
- Deployed across multiple AZs = high availability and fault tolerance.

 **Example:** Example: us-east-1 Region (N. Virginia) has 6 Availability Zones: us-east-1a, us-east-1b, us-east-1c, us-east-1d, us-east-1e, us-east-1f

6.4 Edge Locations & Points of Presence

Edge Locations

Edge Locations are endpoints used by AWS CloudFront (CDN) to cache content closer to the end users. They reduce latency for content delivery.

- 400+ Edge Locations worldwide.
- Separate from Regions and AZs.
- Used by: CloudFront, Route 53, AWS Shield, AWS WAF.

Regional Edge Caches

Larger caches between Edge Locations and AWS Regions — provide a deeper cache layer for content.

6.5 AWS Local Zones

AWS Local Zones are an extension of an AWS Region that places compute, storage, database, and other select AWS services closer to large population centers, industry, and IT centers.

- Designed for applications that require single-digit millisecond latency.
- Use Cases: Media & entertainment, gaming, machine learning inference.

6.6 AWS Wavelength Zones

AWS Wavelength embeds AWS compute and storage services within 5G networks, providing ultra-low latency applications for mobile end-users.

- Use Cases: AR/VR, autonomous vehicles, real-time gaming, smart cities.

6.7 AWS Outposts

AWS Outposts is a fully managed service that extends AWS infrastructure, services, APIs, and tools to customer premises.

- Brings AWS cloud capabilities to your on-premises data center.
- Ideal for workloads requiring local data processing and low latency.

Infrastructure Component	Definition	Scale
Region	Cluster of data centers in a geographic area	30+ worldwide
Availability Zone (AZ)	One or more data centers with independent power/networking within a Region	96+ worldwide
Edge Location	Content delivery endpoint for CloudFront CDN	400+ worldwide
Local Zone	Extension of Region closer to population centers	30+ worldwide
Wavelength Zone	Infrastructure embedded in 5G networks	Select carriers
Outposts	AWS hardware deployed in customer's data center	Customer premises

★ IMPORTANT QUESTIONS FOR UNIVERSITY EXAMS ★

Compiled from JNTUH, Mumbai University, Osmania University, Kathmandu University & AWS Certification Previous Papers

SECTION A: Short Answer Questions (2–4 Marks)

- Q1.** Define Cloud Computing. State any four essential characteristics of cloud computing. [4 Marks]
- Q2.** What is the NIST definition of cloud computing? [2 Marks]
- Q3.** Differentiate between Grid Computing and Cloud Computing. [4 Marks]
- Q4.** What is meant by 'On-demand self-service' in cloud computing? [2 Marks]
- Q5.** Define Elasticity and Scalability in the context of cloud computing. [3 Marks]
- Q6.** What is Multi-tenancy in cloud computing? [2 Marks]
- Q7.** What are the three primary cloud service models? Give one example of each. [3 Marks]
- Q8.** What is AWS? When was it launched? [2 Marks]
- Q9.** What is an AWS Region? How do you choose a Region? [4 Marks]
- Q10.** Define Availability Zones. Why are they important? [3 Marks]
- Q11.** What are Edge Locations in AWS? What service uses them? [2 Marks]
- Q12.** What is the AWS Shared Responsibility Model? [4 Marks]
- Q13.** What is SaaS? Give three real-world examples. [3 Marks]
- Q14.** Distinguish between Public Cloud and Private Cloud. [4 Marks]
- Q15.** What is meant by 'pay-as-you-go' pricing in cloud computing? [2 Marks]

SECTION B: Medium Answer Questions (6–8 Marks)

- Q16.** Explain the six key advantages/benefits of cloud computing with examples. [8 Marks]
- Q17.** Compare IaaS, PaaS, and SaaS with suitable diagrams and examples. Which AWS services fall under each model? [8 Marks]
- Q18.** Explain the four types of cloud deployment models: Public, Private, Hybrid, and Community Cloud with their advantages and disadvantages. [8 Marks]
- Q19.** Describe the AWS Global Infrastructure in detail — Regions, Availability Zones, Edge Locations, and their relationships. [8 Marks]
- Q20.** What is the AWS Shared Responsibility Model? Distinguish between what AWS manages and what the customer manages, with examples. [6 Marks]
- Q21.** Explain Cloud Computing evolution and compare it with traditional on-premise IT

infrastructure. [6 Marks]

Q22. What is High Availability and Fault Tolerance in cloud computing? How does AWS achieve them? [6 Marks]

Q23. Discuss the five essential characteristics of cloud computing as defined by NIST with real examples. [8 Marks]

Q24. Explain Resource Pooling and Measured Service in cloud computing. [6 Marks]

Q25. Compare Cloud Computing with Grid Computing and Utility Computing. [6 Marks]

SECTION C: Long Answer / Essay Questions (10–16 Marks)

Q26. What is cloud computing? Explain in detail the five essential characteristics of cloud computing with examples. Also discuss three service models (IaaS, PaaS, SaaS) and four deployment models (Public, Private, Hybrid, Community) with a comparison table. [16 Marks]

Q27. Explain Amazon Web Services (AWS) in detail. Describe the AWS Global Infrastructure including Regions, Availability Zones, Edge Locations, Local Zones, Wavelength Zones, and Outposts. Why is the global infrastructure important for cloud services? [16 Marks]

Q28. Discuss the advantages and challenges of migrating from on-premises infrastructure to cloud computing. Explain the key factors organizations should consider when choosing between IaaS, PaaS, and SaaS models. [14 Marks]

Q29. Explain the AWS Shared Responsibility Model in detail. What are the security responsibilities of AWS? What responsibilities lie with the customer? Provide a tabular comparison. [10 Marks]

Q30. Write a detailed note on AWS Global Infrastructure. Explain the significance of having multiple Availability Zones in a single Region. How does it contribute to fault tolerance and high availability? [12 Marks]

SECTION D: Multiple Choice Questions (MCQs)

Important MCQs frequently asked in university exams and AWS certification examinations:

Q1. Which of the following is NOT an essential characteristic of cloud computing according to NIST?

- a) On-demand self-service
- b) Broad network access
- c) Dedicated hardware per user
- d) Measured service

✓ **Answer: c) Dedicated hardware per user**

Q2. Amazon EC2 (Elastic Compute Cloud) falls under which cloud service model?

- a) SaaS
- b) PaaS
- c) IaaS
- d) FaaS

✓ **Answer: c) IaaS**

Q3. In which cloud service model does the customer only manage Applications and Data?

- a) IaaS
- b) PaaS
- c) SaaS
- d) On-premises

✓ **Answer: b) PaaS**

Q4. Which AWS Global Infrastructure component is one or more discrete data centers with redundant power, networking, and connectivity?

- a) Edge Location
- b) Region
- c) Availability Zone
- d) Local Zone

✓ **Answer: c) Availability Zone**

Q5. What is the primary pricing model for AWS services?

- a) Pay a fixed monthly fee
- b) Pay-as-you-go
- c) Annual subscription only
- d) Free forever

✓ **Answer: b) Pay-as-you-go**

Q6. AWS CloudFront (CDN) uses which infrastructure component to cache content closer to users?

- a) Availability Zones
- b) Regions
- c) Edge Locations
- d) Local Zones

✓ **Answer: c) Edge Locations**

Q7. Which cloud deployment model is used by a specific group of organizations with common concerns (e.g., security, compliance)?

- a) Public Cloud
- b) Private Cloud
- c) Hybrid Cloud
- d) Community Cloud

✓ **Answer: d) Community Cloud**

Q8. Which of the following is an example of SaaS?

- a) Amazon EC2
- b) AWS Elastic Beanstalk
- c) Gmail / Microsoft Office 365
- d) Amazon S3

✓ **Answer: c) Gmail / Microsoft Office 365**

Q9. In the AWS Shared Responsibility Model, who is responsible for patching the operating system on an EC2 instance?

- a) AWS
- b) Customer
- c) Both equally
- d) Neither

✓ **Answer: b) Customer**

Q10. Which of the following AWS services is used for managed relational databases?

- a) Amazon DynamoDB
- b) Amazon RDS
- c) Amazon S3
- d) AWS Lambda

✓ **Answer: b) Amazon RDS**

Q11. What does 'Elasticity' mean in cloud computing?

- a) The ability to only scale upward
- b) The ability to automatically acquire and release resources based on demand
- c) Using elastic bands in servers
- d) Charging customers more during peak times

✓ **Answer: b) The ability to automatically acquire and release resources based on demand**

Q12. AWS Outposts is designed to:

- a) Extend AWS services to customer's on-premises data center
- b) Provide 5G mobile services
- c) Replace all physical data centers
- d) Offer free cloud services

✓ **Answer: a) Extend AWS services to customer's on-premises data center**

Q13. Which factor is NOT typically considered when choosing an AWS Region?

- a) Proximity to customers
- b) Compliance requirements
- c) Server brand/manufacture
- d) Pricing

✓ **Answer: c) Server brand/manufacture**

Q14. How many minimum Availability Zones does each AWS Region have?

- a) 1
- b) 2
- c) 5
- d) 10

✓ **Answer: b) 2**

Q15. Which service is an example of PaaS on AWS?

- a) Amazon EC2
- b) AWS Elastic Beanstalk
- c) Amazon S3
- d) Amazon CloudFront

✓ **Answer: b) AWS Elastic Beanstalk**

SECTION E: Fill in the Blanks

1. Cloud computing delivers IT resources over the _____ (internet/cloud).
2. The three primary cloud service models are _____, _____, and _____.
3. AWS stands for _____.
4. An AWS _____ is a physical location in the world where AWS clusters data centers.
5. _____ Zones are one or more discrete data centers within a Region with independent power and networking.
6. CloudFront uses _____ to cache content closer to users globally.
7. The NIST definition lists _____ essential characteristics of cloud computing.
8. In IaaS, the customer manages the _____ while the cloud provider manages the _____.
9. The _____ cloud model combines public and private cloud environments.
10. AWS was launched in the year _____ with services like S3 and EC2.

✓ **Answers:** 1. Internet 2. IaaS, PaaS, SaaS 3. Amazon Web Services 4. Region 5. Availability 6. Edge Locations 7. Five (5) 8. OS/Application; Hardware/Network 9. Hybrid 10. 2006

SECTION F: True or False Questions

1. Cloud computing always requires the customer to own physical hardware. [FALSE]
2. AWS Global Cloud footprint enables you to deploy applications around the world. [TRUE]
3. In SaaS, the customer is responsible for managing the underlying OS. [FALSE]
4. An AWS Region can have only one Availability Zone. [FALSE]
5. Edge Locations are used by AWS CloudFront for content delivery. [TRUE]
6. IaaS offers the highest level of control and flexibility to customers. [TRUE]
7. In Public Cloud, the infrastructure is dedicated to a single organization. [FALSE]
8. AWS Outposts brings AWS services to on-premises data centers. [TRUE]
9. Rapid elasticity means resources can be scaled quickly but not reduced. [FALSE]
10. Amazon RDS is an example of a Platform as a Service (PaaS) offering. [TRUE]

QUICK REVISION CHEAT SHEET

Key Acronyms & Full Forms

Acronym	Full Form	Context
AWS	Amazon Web Services	Cloud platform by Amazon
IaaS	Infrastructure as a Service	Cloud service model
PaaS	Platform as a Service	Cloud service model
SaaS	Software as a Service	Cloud service model
FaaS	Function as a Service	Serverless computing
NIST	National Institute of Standards and Technology	Defines cloud computing standards
AZ	Availability Zone	AWS infrastructure component
CDN	Content Delivery Network	Used by CloudFront / Edge Locations
VPC	Virtual Private Cloud	AWS networking service
IAM	Identity and Access Management	AWS security service
EC2	Elastic Compute Cloud	AWS virtual server service
S3	Simple Storage Service	AWS object storage
RDS	Relational Database Service	AWS managed database
WAF	Web Application Firewall	AWS security service
DDoS	Distributed Denial of Service	Type of cyber attack; AWS Shield protects against it

Critical Numbers to Remember

- **AWS Regions:** 30+ worldwide
- **Availability Zones:** 96+ worldwide (min 2–3 per Region)
- **Edge Locations:** 400+ worldwide
- **AWS Services:** 200+ services
- **NIST Characteristics:** 5 essential characteristics
- **Cloud Service Models:** 3 primary (IaaS, PaaS, SaaS)
- **Cloud Deployment Models:** 4 types (Public, Private, Hybrid, Community)
- **AWS Founded:** 2006
- **AWS Market Share:** #1 globally (~32%)